Laboratory Report:
I–V Characteristics and Efficiency of a Solar Panel

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Abstract

This report analyses measured I–V data of a small solar panel at three illumination distances: 60 cm, 70 cm and 80 cm. Using the supplied thermopile calibration and measured I–V points, we compute the irradiance, incident optical power on the panel, electrical powers, maximum power points, fill-factor (FF), and conversion efficiency (η). A complete uncertainty analysis using first-order error propagation is included. All intermediate arithmetic steps are shown explicitly and numerical uncertainties are reported.

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1 Theory

A solar cell is a semiconductor device that converts incident light into electrical energy through the photovoltaic effect. When photons with energy greater than the bandgap are absorbed, electron-hole pairs are generated. These charge carriers are separated by the internal electric field of the p–n junction, leading to a photocurrent.

Equivalent circuit and I–V characteristic

The simplest equivalent circuit of an illuminated solar cell consists of:

1. a current source representing the light-generated current I_L,
2. a diode representing the p–n junction,
3. a series resistance R_s (contact and bulk resistance),
4. and a shunt resistance R_{sh} (leakage paths).

The total current is expressed as:

$$I = I_L - I_0\left(e^{\frac{qV}{nk_BT}} - 1\right) - \frac{V + IR_s}{R_{sh}}, \tag{1}$$

where I_0 is the dark saturation current, q the electronic charge, n the ideality factor, k_B Boltzmann's constant, and T the absolute temperature. In most laboratory conditions, R_{sh} is large and can be neglected, simplifying the model.

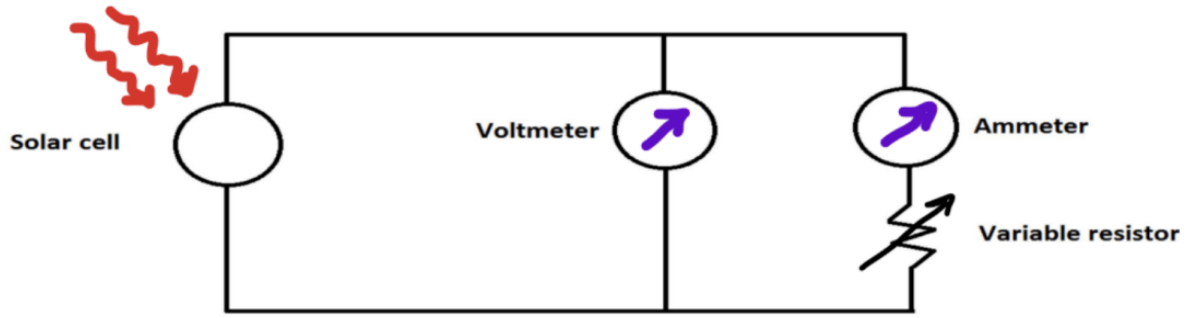


Figure 1: Schematic circuit diagram

Important parameters

From the I–V curve of an illuminated solar cell:

- **Short-circuit current** I_{SC}: the current when the output voltage is zero ($V = 0$). It is proportional to the light intensity incident on the cell.
- **Open-circuit voltage** V_{OC}: the terminal voltage when the output current is zero ($I = 0$). It increases logarithmically with light intensity.
- **Maximum power point (MPP)**: the operating point (V_m, I_m) where the product

$$P_m = V_m I_m \tag{2}$$

is maximum. This point represents the highest electrical power the cell can deliver under the given illumination.

- **Fill factor (FF)**: a measure of the "squareness" of the I–V curve, defined as

$$\mathrm{FF} = \frac{P_m}{V_{OC} I_{SC}} = \frac{V_m I_m}{V_{OC} I_{SC}}. \tag{3}$$

  Higher FF values (close to unity) indicate lower internal losses.

- **Conversion efficiency** η: the ratio of maximum electrical output power to the incident optical power on the panel,

$$\eta = \frac{P_m}{P_{\mathrm{in}}} = \frac{\mathrm{FF}\, V_{OC} I_{SC}}{P_{\mathrm{in}}}. \tag{4}$$

Incident power calculation

The incident optical power on the solar panel is obtained from:

$$P_{\mathrm{in}} = J \times A, \tag{5}$$

where J is the irradiance (in $\mathrm{W/m^2}$) measured by a calibrated thermopile sensor, and A is the active area of the panel. In this experiment, irradiance was determined using the calibration constant of the thermopile, and the panel area was known.

Summary

Thus, by experimentally measuring the I–V curve of the solar panel under different illumination intensities, one can determine the characteristic parameters $(I_{SC}, V_{OC}, P_m, \mathrm{FF}, \eta)$. These parameters are essential to evaluate the performance of solar cells.

Key performance metrics are:

- Short-circuit current I_{SC},
- Open-circuit voltage V_{OC},
- Maximum power $P_m = V_m I_m$ at (V_m, I_m),
- Fill factor $\mathrm{FF} = \dfrac{P_m}{V_{OC} I_{SC}}$,
- Conversion efficiency $\eta = \dfrac{P_m}{P_{\mathrm{in}}}$.

This report calculates the above for measured data, with detailed uncertainty propagation.

2 Experimental details and constants used

- Thermopile calibration factor (given): $k = 40.43\ \mathrm{W\,m^{-2}\,mV^{-1}}$. Thus irradiance $J = k \cdot V_{\text{thermopile (mV)}}$.
- Active area of the solar panel (given): $A = 9.0 \times 10^{-3}\ \mathrm{m^2}$.
- Units: currents are recorded in mA; for power calculations we use $P(\mathrm{mW}) = V(\mathrm{V}) \times I(\mathrm{mA})$.

3 Assumed instrument uncertainties (used in error analysis)

(These are stated here explicitly so the reader knows how propagated uncertainties were computed.)

- Thermopile calibration constant k: relative uncertainty $u_k/k = 1.0\%$.
- Thermopile voltage reading V_t (mV): absolute uncertainty $u_{V_t} = 0.2$ mV (repeatability / reading resolution).
- Panel area A: relative uncertainty $u_A/A = 1.0\%$.
- Voltmeter (measuring cell voltage V): modelled as $u_V = \sqrt{(0.5\% \times V)^2 + (0.001\ \text{V})^2}$.
- Ammeter (measuring current I in mA): modelled as $u_I = \sqrt{(0.5\% \times I)^2 + (0.1\ \text{mA})^2}$.

All uncertainty propagation below uses first-order (linear) propagation (root-sum-of-squares) appropriate for small relative uncertainties.

4 Raw measured quantities

Below are the thermopile readings (mV) and the representative measured I–V rows used to compute P. For each distance the maximum listed P is taken as P_m; its corresponding (V_m, I_m) is used.

Distance = 60 cm

Thermopile reading: $V_t = 36.9$ mV.

Representative I–V points (Voltage V in V, Current I in mA):

V (V)	I (mA)	$P = V \cdot I$ (mW)
0.54	101.5	54.81
0.71	136.0	96.56
0.99	137.0	135.63
1.10	136.0	149.60
1.48	138.0	204.24
1.74	138.5	240.99
1.80	136.4	245.52
1.82	137.3	**249.886**
1.84	133.6	245.824

Table 1: Selected measured points (60 cm). Maximum listed $P_m = 249.886$ mW at $V_m = 1.82$ V, $I_m = 137.3$ mA.

Distance = 70 cm

Thermopile reading: $V_t = 24.9$ mV.

V (V)	I (mA)	P (mW)
1.31	103.0	134.93
1.57	103.5	162.495
1.63	103.6	168.868
1.73	101.8	176.114
1.86	97.9	**182.094**
1.90	93.4	177.46

Table 2: Selected measured points (70 cm). Maximum listed $P_m = 182.094$ mW at $V_m = 1.86$ V, $I_m = 97.9$ mA.

Distance = 80 cm

Thermopile reading: $V_t = 18.5$ mV.

V (V)	I (mA)	P (mW)
1.24	103.3	128.092
1.31	103.0	134.93
1.76	76.9	135.344
1.81	76.8	**139.008**
1.86	73.4	136.524

Table 3: elected measured points (80 cm). Maximum listed $P_m = 139.008$ mW at $V_m = 1.81$ V, $I_m = 76.8$ mA.

5 Plots

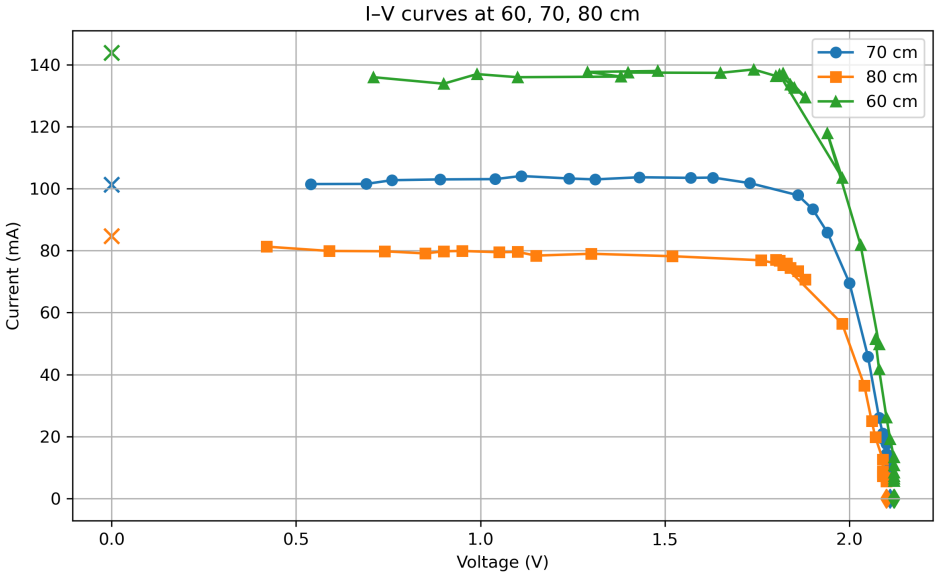


Figure 2: I-V characteristics of the solar cell for different intensities

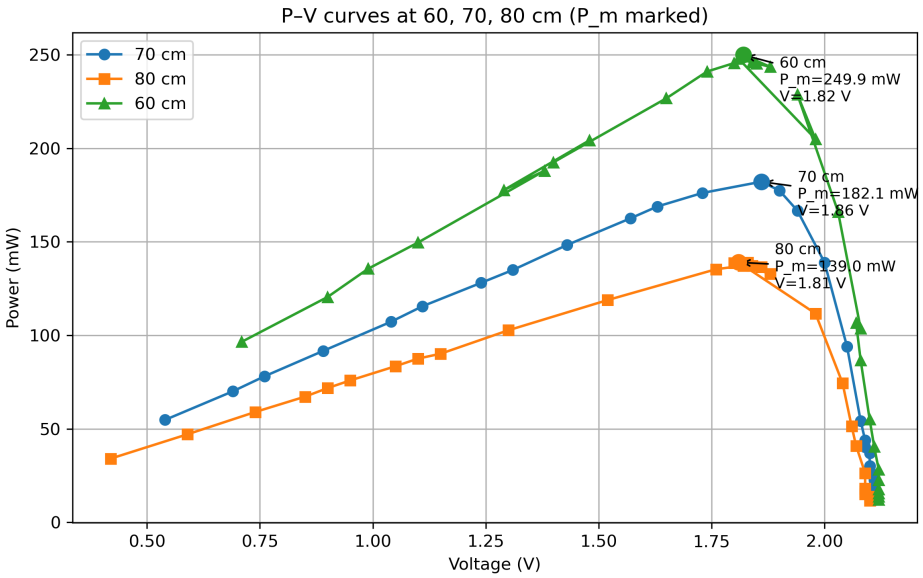


Figure 3: Power vs. Voltage plot for solar cell for different intensities

6 Step-by-step calculations (numeric)

Below we calculate irradiance J, incident power P_{in}, then compute FF and η for each distance. All arithmetic is shown.

Formulas

$$J = k \cdot V_t \quad (\text{W/m}^2), \tag{6}$$

$$P_{\text{in}} = J \cdot A \quad (\text{W}) = JA \quad \text{– to convert to mW multiply by 1000.} \tag{7}$$

$$P = V \cdot I \quad (\text{mW; with } I \text{ in mA}), \tag{8}$$

$$\text{FF} = \frac{P_m}{V_{OC}\, I_{SC}} \quad (\text{dimensionless, with } I_{SC} \text{ in mA gives } V \cdot I \text{ in mW}), \tag{9}$$

$$\eta = \frac{P_m}{P_{\text{in}}}. \tag{10}$$

Distance 60 cm – numeric

$$V_t = 36.9 \text{ mV},$$
$$J = 40.43 \times 36.9 = 1491.867 \text{ W/m}^2,$$
$$P_{\text{in}} = J \times A = 1491.867 \times 9 \times 10^{-3} = 13.426803 \text{ W} = 13426.803 \text{ mW}.$$

Measured: $V_{OC} = 2.17$ V, $I_{SC} = 136.65$ mA. Maximum electrical power (measured): $P_m = 249.886$ mW at $V_m = 1.82$ V, $I_m = 137.3$ mA.

$$\text{FF} = \frac{249.886}{2.17 \times 136.65} = \frac{249.886}{296.5905} = 0.8427.$$

$$\eta = \frac{249.886}{13426.803} = 0.01861 = 1.861\%.$$

Distance 70 cm – numeric

$$\begin{aligned} V_t &= 24.9 \text{ mV}, \\ J &= 40.43 \times 24.9 = 1006.707 \text{ W/m}^2, \\ P_{\text{in}} &= 1006.707 \times 9 \times 10^{-3} = 9.060363 \text{ W} = 9060.363 \text{ mW}. \end{aligned}$$

Measured: $V_{OC} = 2.135$ V, $I_{SC} = 101.8$ mA. $P_m = 182.094$ mW at $V_m = 1.86$ V, $I_m = 97.9$ mA.

$$\text{FF} = \frac{182.094}{2.135 \times 101.8} = \frac{182.094}{217.943} = 0.8378.$$

$$\eta = \frac{182.094}{9060.363} = 0.02008 = 2.008\%.$$

Distance 80 cm – numeric

$$\begin{aligned} V_t &= 18.5 \text{ mV}, \\ J &= 40.43 \times 18.5 = 747.955 \text{ W/m}^2, \\ P_{\text{in}} &= J \times A = 747.955 \times 9 \times 10^{-3} = 6.731595 \text{ W} = 6731.595 \text{ mW}. \end{aligned}$$

Measured: $V_{OC} = 2.11$ V, $I_{SC} = 79.75$ mA. The maximum measured electrical power is

$$P_m = 139.008 \text{ mW}$$

occurring at $V_m = 1.81$ V, $I_m = 76.8$ mA.

$$\text{FF} = \frac{P_m}{V_{OC}\, I_{SC}} = \frac{139.008}{2.11 \times 79.75} = \frac{139.008}{168.2225} = 0.82609.$$

$$\eta = \frac{P_m}{P_{\text{in}}} = \frac{139.008}{6731.595} = 0.02065008 = 2.0650\%.$$

7 Uncertainty (error) analysis

We propagate uncertainties using first-order formulas. Relevant propagation rules used:

- For $X = ab\cdots$ (product), relative variance: $\left(\frac{u_X}{X}\right)^2 = \left(\frac{u_a}{a}\right)^2 + \left(\frac{u_b}{b}\right)^2 + \cdots$ (assuming independent errors).
- For $P = V \cdot I$, where I is in mA and V in V so P in mW:

$$u_P = \sqrt{(I\, u_V)^2 + (V\, u_I)^2}.$$

- For ratios such as $\text{FF} = P_m/(V_{OC} I_{SC})$:

$$\left(\frac{u_{\text{FF}}}{\text{FF}}\right)^2 = \left(\frac{u_{P_m}}{P_m}\right)^2 + \left(\frac{u_{V_{OC}}}{V_{OC}}\right)^2 + \left(\frac{u_{I_{SC}}}{I_{SC}}\right)^2.$$

- For $\eta = P_m/P_{\text{in}}$, similarly

$$\left(\frac{u_\eta}{\eta}\right)^2 = \left(\frac{u_{P_m}}{P_m}\right)^2 + \left(\frac{u_{P_{\text{in}}}}{P_{\text{in}}}\right)^2.$$

We now compute uncertainties numerically using the instrument uncertainty model stated above.

Distance 60 cm

Incident power uncertainty

$$u_k/k = 1\% = 0.010, \quad u_{V_t} = 0.2 \text{ mV}, \quad V_t = 36.9 \text{ mV}, \quad u_A/A = 1\% = 0.010.$$

Relative uncertainty of $P_{\text{in}} = kV_tA$:

$$\left(\frac{u_{P_{\text{in}}}}{P_{\text{in}}}\right) = \sqrt{(0.010)^2 + \left(\frac{0.2}{36.9}\right)^2 + (0.010)^2} = \sqrt{0.000100 + 0.0000294 + 0.000100} = 0.01514.$$

Thus

$$u_{P_{\text{in}}} = 0.01514 \times 13426.803 = 203.4 \text{ mW}.$$

Maximum measured power uncertainty Using $V_m = 1.82$ V, $I_m = 137.3$ mA.

Voltmeter uncertainty:

$$u_{V_m} = \sqrt{(0.005 \times 1.82)^2 + (0.001)^2} = 0.009154 \text{ V}.$$

Ammeter uncertainty:

$$u_{I_m} = \sqrt{(0.005 \times 137.3)^2 + (0.1)^2} = 0.6938 \text{ mA}.$$

Now

$$u_{P_m} = \sqrt{(I_m u_{V_m})^2 + (V_m u_{I_m})^2} = \sqrt{(137.3 \times 0.009154)^2 + (1.82 \times 0.6938)^2} = 1.782 \text{ mW}.$$

FF uncertainty We need $u_{V_{OC}}$ and $u_{I_{SC}}$:

$$u_{V_{OC}} = \sqrt{(0.005 \times 2.17)^2 + 0.001^2} = 0.01089 \text{ V}, \quad u_{I_{SC}} = \sqrt{(0.005 \times 136.65)^2 + 0.1^2} = 0.6913 \text{ mA}.$$

Relative uncertainty of FF:

$$\frac{u_{\text{FF}}}{\text{FF}} = \sqrt{(1.782/249.886)^2 + (0.01089/2.17)^2 + (0.6913/136.65)^2} = 0.01003.$$

So

$$u_{\text{FF}} = 0.01003 \times 0.8427 = 0.0085.$$

Efficiency uncertainty Relative uncertainty:

$$\frac{u_\eta}{\eta} = \sqrt{(1.782/249.886)^2 + (203.4/13426.803)^2} = 0.01679.$$

So

$$u_\eta = 0.01679 \times 0.01861 = 0.000313 \quad \Rightarrow \quad u_\eta = 0.0313\% \text{ points}.$$

Distance 60 cm – reported values (with uncertainties)

$$P_{\text{in}} = 13426.8 \pm 203.4 \text{ mW}, \quad P_m = 249.886 \pm 1.782 \text{ mW},$$

$$\text{FF} = 0.8427 \pm 0.0085, \quad \eta = 1.861\% \pm 0.031\% \text{ (absolute percentage points)}.$$

Distance 70 cm

(Concise numeric steps; same procedures as above.)

Incident power

$$V_t = 24.9 \text{ mV}, \ J = 1006.707 \text{ W/m}^2, \ P_{\text{in}} = 9060.363 \text{ mW}.$$

Relative uncertainty:

$$\frac{u_{P_{\text{in}}}}{P_{\text{in}}} = \sqrt{(0.010)^2 + (0.2/24.9)^2 + (0.010)^2} = 0.01627,$$

so

$$u_{P_{\text{in}}} = 0.01627 \times 9060.363 = 147.5 \text{ mW}.$$

Maximum measured power At $V_m = 1.86$ V, $I_m = 97.9$ mA:

$$u_{V_m} = 0.009357 \text{ V}, \quad u_{I_m} = 0.4996 \text{ mA},$$

$$u_{P_m} = \sqrt{(97.9 \times 0.009357)^2 + (1.86 \times 0.4996)^2} = 1.304 \text{ mW}.$$

FF and η uncertainties

$$u_{V_{OC}} = 0.01072 \text{ V}, \quad u_{I_{SC}} = 0.5188 \text{ mA}.$$

Relative uncertainty of FF: 0.010025 so

$$\text{FF} = 0.8378 \pm 0.0084.$$

Relative uncertainty of η: 0.01778 so

$$\eta = 2.008\% \pm 0.036\%.$$

Distance 80 cm

$$V_t = 18.5 \text{ mV}, \quad J = 747.955 \text{ W/m}^2, \quad P_{\text{in}} = 6731.595 \text{ mW}.$$

Relative uncertainty of P_{in} (using $u_k/k = 1\%$, $u_{V_t} = 0.2$ mV, $u_A/A = 1\%$):

$$\frac{u_{P_{\text{in}}}}{P_{\text{in}}} = \sqrt{(0.010)^2 + \left(\frac{0.2}{18.5}\right)^2 + (0.010)^2} = 0.01780,$$

so

$$u_{P_{\text{in}}} = 0.01780 \times 6731.595 = 119.8 \text{ mW}.$$

Maximum measured power At $V_m = 1.81$ V, $I_m = 76.8$ mA:

$$u_{V_m} = \sqrt{(0.005 \times 1.81)^2 + (0.001)^2} = 0.00911 \text{ V},$$

$$u_{I_m} = \sqrt{(0.005 \times 76.8)^2 + (0.1)^2} = 0.3968 \text{ mA}.$$

Thus the uncertainty on P_m is

$$u_{P_m} = \sqrt{(I_m \, u_{V_m})^2 + (V_m \, u_{I_m})^2} \approx 1.002 \text{ mW}.$$

FF and η uncertainties For $V_{OC} = 2.11$ V and $I_{SC} = 79.75$ mA we have

$$u_{V_{OC}} \approx 0.01060 \text{ V}, \quad u_{I_{SC}} \approx 0.411 \text{ mA}.$$

Propagating errors,

$$\frac{u_{\text{FF}}}{\text{FF}} = \sqrt{\left(\frac{u_{P_m}}{P_m}\right)^2 + \left(\frac{u_{V_{OC}}}{V_{OC}}\right)^2 + \left(\frac{u_{I_{SC}}}{I_{SC}}\right)^2}$$

gives

$$\text{FF} = 0.82609 \pm 0.00842.$$

For efficiency,

$$\frac{u_\eta}{\eta} = \sqrt{\left(\frac{u_{P_m}}{P_m}\right)^2 + \left(\frac{u_{P_{\text{in}}}}{P_{\text{in}}}\right)^2},$$

so

$$\eta = 2.0650\% \pm 0.0397\% \text{ (absolute percentage points)}.$$

8 Summary Table (final results)

Distance	P_{in} (mW)	P_m (mW)	FF	η (%)
60 cm	13426.8 ± 203.4	249.886 ± 1.782	0.8427 ± 0.0085	1.861 ± 0.031
70 cm	9060.36 ± 147.5	182.094 ± 1.304	0.8378 ± 0.0084	2.008 ± 0.036
80 cm	6731.60 ± 119.8	139.008 ± 1.002	0.82609 ± 0.00842	2.065 ± 0.040

Table 4: Final computed results with propagated uncertainties (absolute errors shown).

9 Discussion

- The absolute conversion efficiencies ($\approx 1.9\% - 2.0\%$) are low relative to commercial modules (10–22%) because:
  - small lab cells and non-standard lamp spectra (not AM1.5) cause spectral mismatch,
  - geometric/angle mismatch and non-uniform illumination,
  - measurement uncertainties and limited area; and
  - possible heating and temperature effects reducing V_{OC}.
- Uncertainty budget: the major contributor to η uncertainty is the incident power uncertainty (thermopile calibration + area), while the major contributor to FF uncertainty is the electrical measurement uncertainty near the maximum power point (voltmeter/ammeter).

10 Conclusions

A full I–V analysis was performed for three illumination distances. The maximum power points were identified directly from measured P values; FF and η were computed and uncertainties propagated. All intermediate numeric steps and error calculations are reported above for reproducibility.

11 Appendix

Full Data Sets (I-V-Power measurement table for different distances):

for 60cm		
V	**Current(mA)**	**power(mW)**
0.71	136	96.56
0.9	133.9	120.51
0.99	137	135.63
1.1	136	149.6
1.38	136.2	187.956
1.29	137.7	177.633
1.48	138	204.24
1.4	137.5	192.5
1.65	137.4	226.71
1.74	138.5	240.99
1.8	136.4	245.52
1.82	137.3	249.886
1.88	129.5	243.46
1.85	132.6	245.31
1.84	133.6	245.824
1.81	136.9	247.789
1.98	103.5	204.93
1.94	117.9	228.726
2.03	81.8	166.054
2.08	49.8	103.584
2.07	51.5	106.605
2.08	41.7	86.736
2.1	26.2	55.02
2.11	19.2	40.512
2.12	13.3	28.196
2.12	10.7	22.684
2.12	8.4	17.808
2.12	7.3	15.476
2.12	6.5	13.78
2.12	5.7	12.084

for 70cm		
V	**Current(mA)**	**power(mW)**
0.54	101.5	54.81
0.69	101.55	70.0695
0.76	102.75	78.09
0.89	103	91.67
1.04	103.1	107.224
1.11	104.1	115.551
1.24	103.3	128.092
1.31	103	134.93
1.43	103.7	148.291
1.57	103.5	162.495
1.63	103.6	168.868
1.73	101.8	176.114
1.86	97.9	182.094
1.9	93.4	177.46
1.94	85.9	166.646
2	69.5	139
2.05	45.8	93.89
2.08	26.1	54.288
2.09	21	43.89
2.09	19.3	40.337
2.1	17.6	36.96
2.1	14.4	30.24
2.1	13	27.3
2.11	10.7	22.577
2.11	9	18.99
2.11	7.9	16.669
2.11	7	14.77
2.11	6.5	13.715
2.11	5.6	11.816

for 80cm		
voltage(V)	**Current(mA)**	**power(mW)**
0.42	81.3	34.146
0.59	79.9	47.141
0.74	79.8	59.052
0.85	79.1	67.235
0.9	79.8	71.82
0.95	79.9	75.905
1.05	79.5	83.475
1.1	79.6	87.56
1.15	78.4	90.16
1.3	79	102.7
1.52	78.2	118.864
1.76	76.9	135.344
1.82	75.4	137.228
1.88	70.7	132.916
1.84	74.5	137.08
1.86	73.4	136.524
1.8	77	138.6
1.83	75.9	138.897
1.81	76.8	139.008
1.98	56.4	111.672
2.04	36.5	74.46
2.06	25	51.5
2.07	19.8	40.986
2.09	12.6	26.334
2.09	8.7	18.183
2.09	7.9	16.511
2.09	7.2	15.048
2.1	6.6	13.86
2.1	5.7	11.97
2.1	5.6	11.76